

Measuring The Effects of Mental Fatigue on Worker Performance Through Task Rotations

How does rotation over tasks with different characteristics effect the impacts of mental fatigue on worker performance?

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Background and Motivation

- Mental fatigue** is a key factor that affects efficiency and accuracy of work. Task characteristics, such as **cognitive workload** and **task complexity**, play a significant role in how mental fatigue develops during work.
- High levels of fatigue has **serious consequences** for worker **safety and health** (CDC, 2024).
- Workers with fatigue may have **reduced reaction time**, **lower productivity** levels, and a **reduction** in the ability for **complex planning** (Cunningham et. Al., 2022).

Task rotations are widely used to manage both physical and mental fatigue. Our study investigates the effects of task rotations on mental fatigue through the usage of:



Task Completion Time



Errors Made



Self-assessment

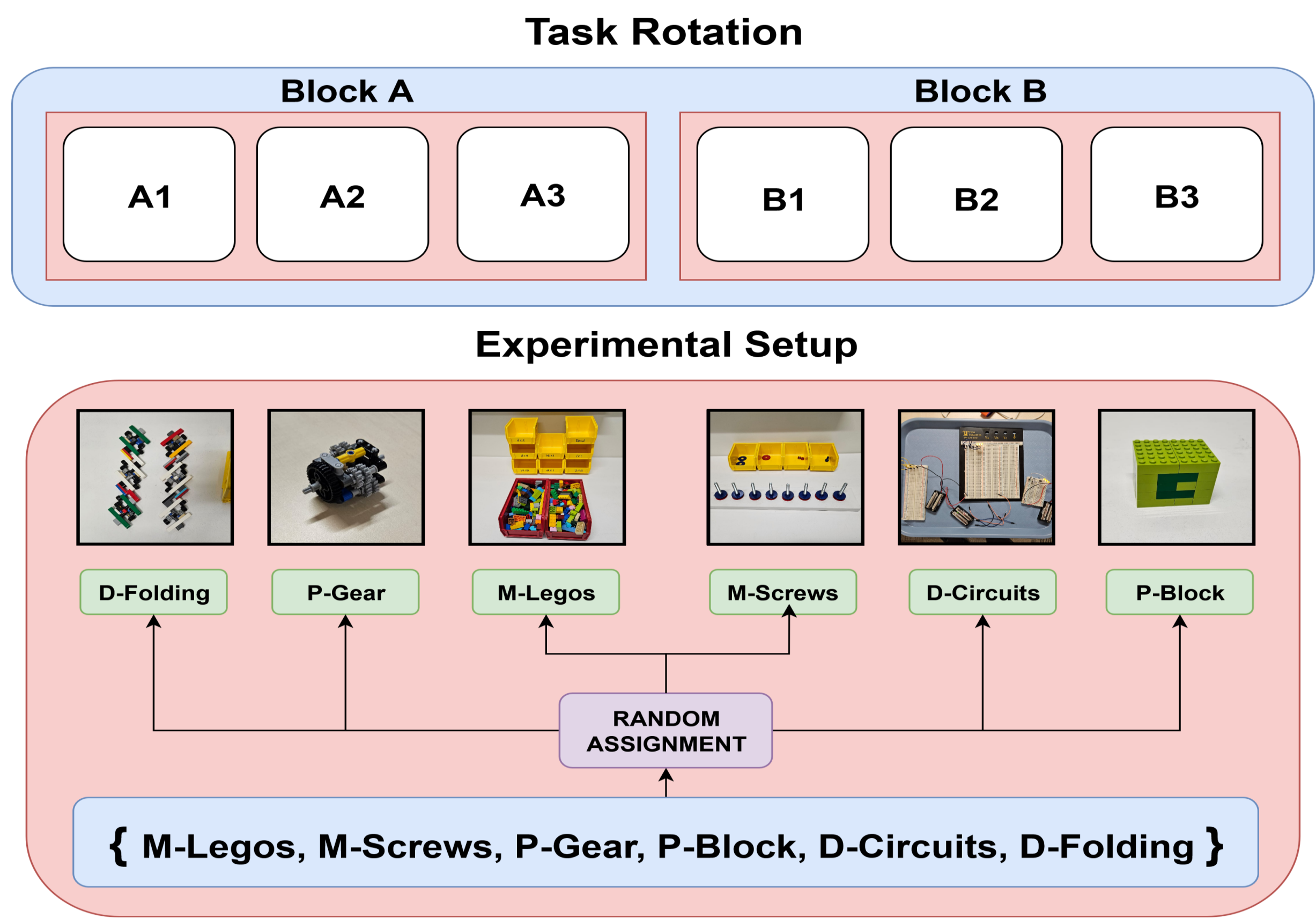
Experimental Setup

- Fatigue was induced in participants through **non-stop** work.
- After each task, the participant will self-assess their fatigue through the NASA TLX and a custom-made questionnaire.
- Participant's hand movements are captured with camera.



Experimental Design

Experiment Tasks		
Diagnostic	Procedural	Manual
Physical Effort: Low Mental Effort: High	Physical Effort: Low–Moderate Mental Effort: Moderate	Physical Effort: Moderate-High Mental Effort: Low–Moderate
D-Folding	P-Gear	M-Screw
The participant is required to find and fix the error in a LEGO build restore the functionality of the assembly.	This task requires participants to construct a functional gear by following a detailed instruction manual.	Participants are tasked with screwing washers and bolts onto a pegboard in a specified order given in the instructions.
D-Circuit	P-Block	M-Lego
The participant is required to troubleshoot 2 circuits in order to illuminate 3 LEDs.	This task requires participants to construct a LEGO cube featuring the letters C, I, T, and C by following a detailed instruction manual.	Participants are tasked with sorting a variety of loose LEGO pieces into 8 bins based on the size and shape of the bricks.

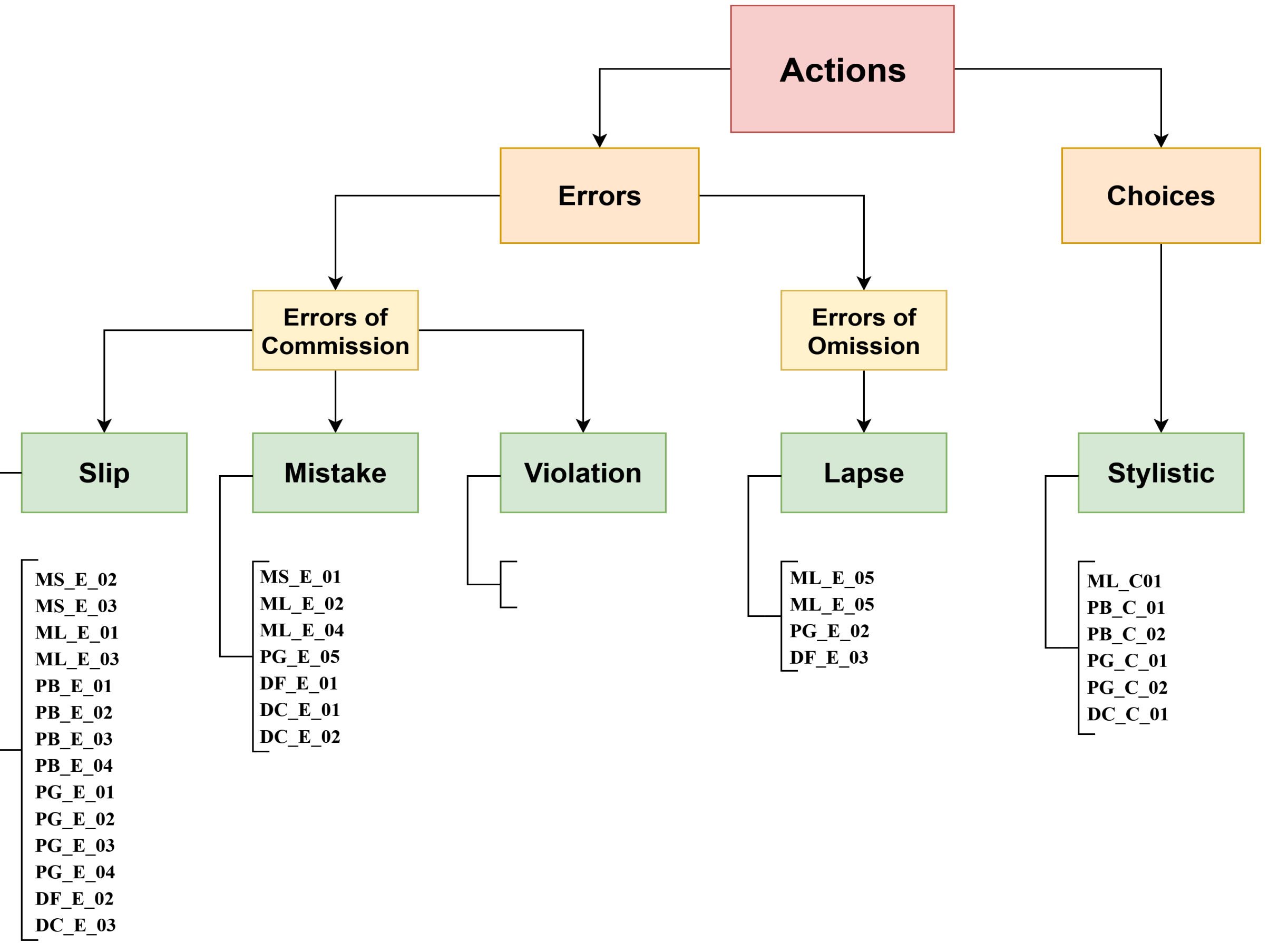


Error Coding

We proposed a set of specifically defined actions that can be observed in the video recordings and contrasted amongst participants.

Error types were categorized to determine the frequency and severity of each type relative to participant fatigue levels.

Error Taxonomy



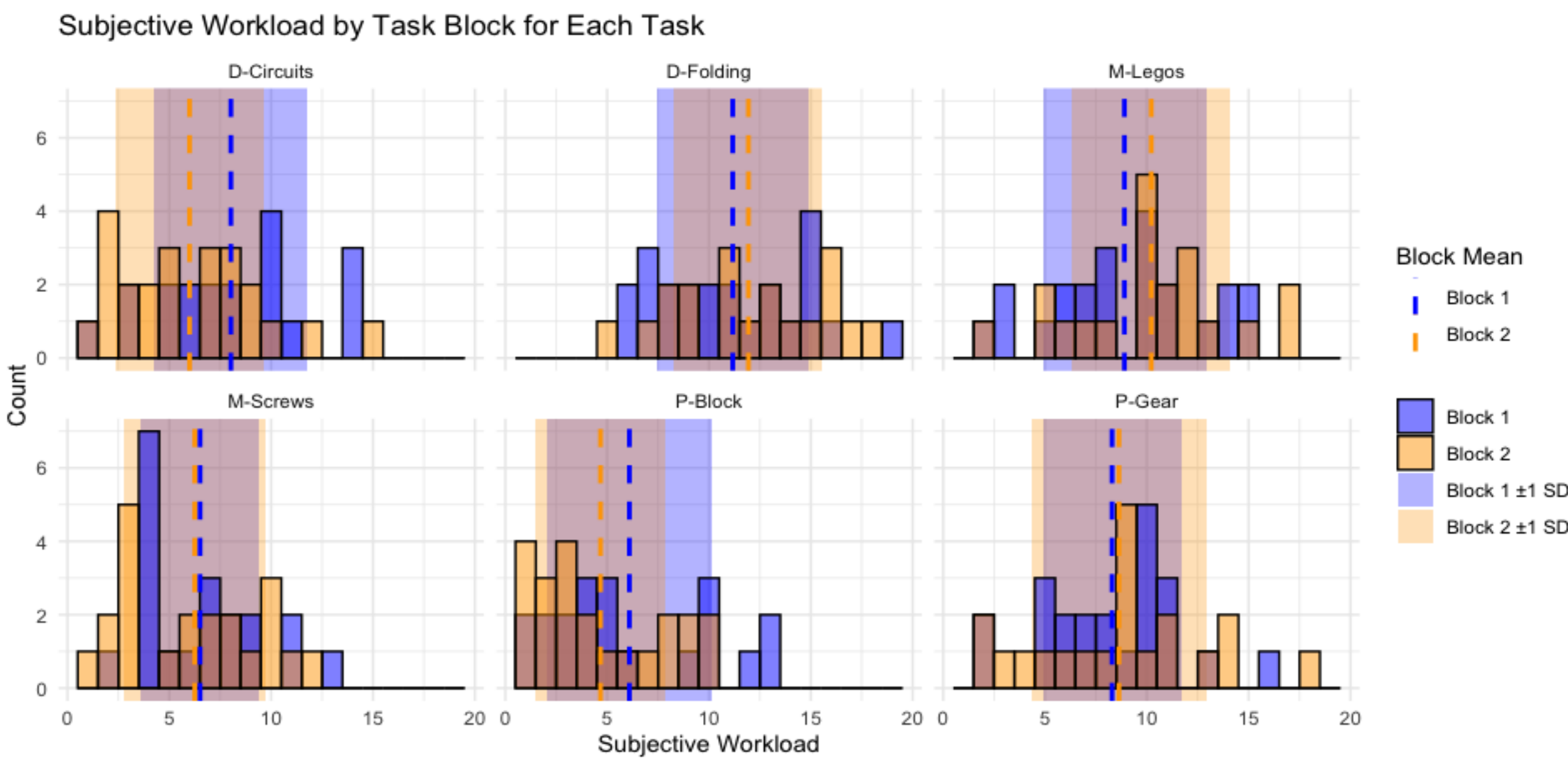
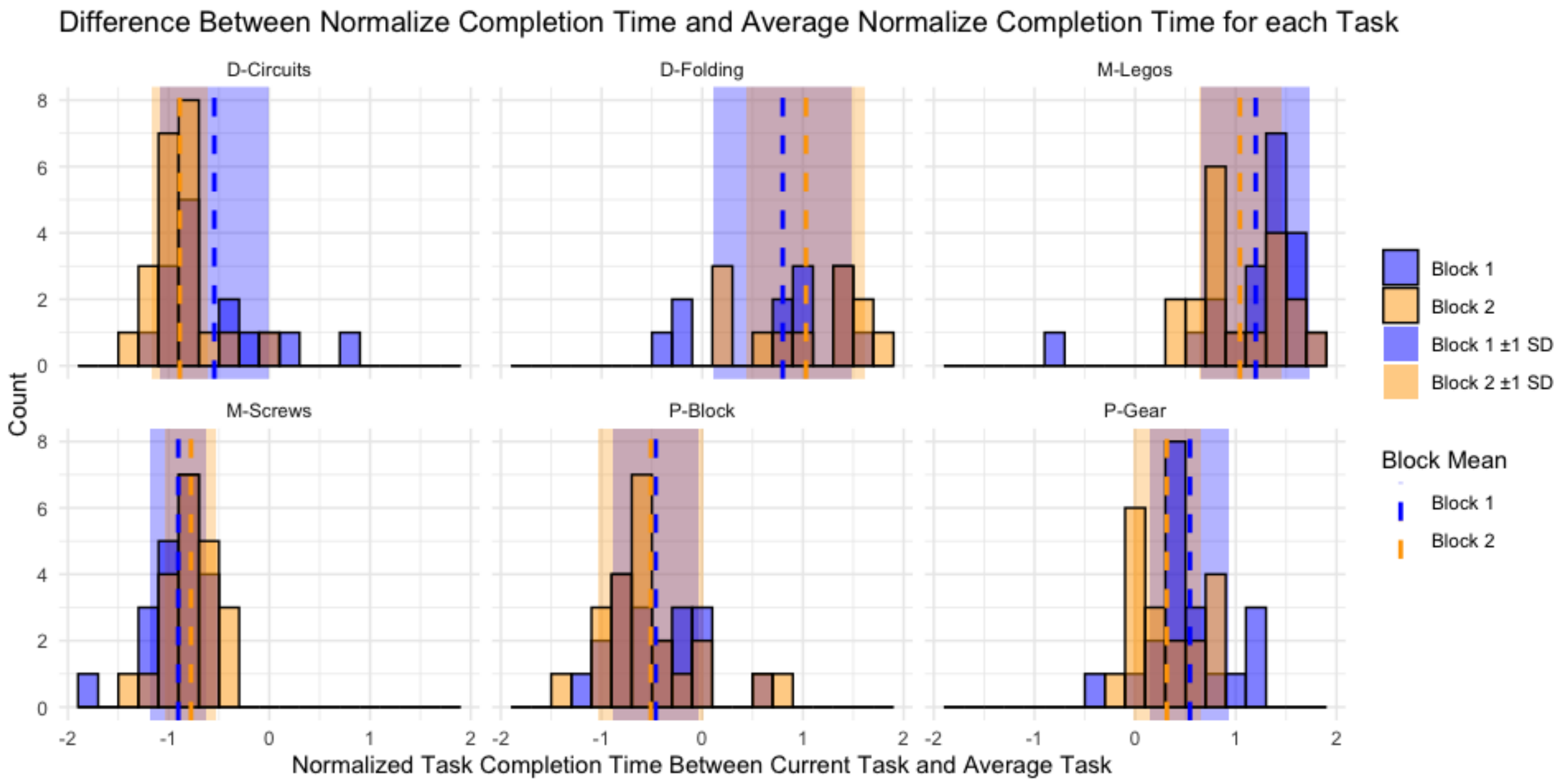
Sample Demographics

The data was collected from 48 college-aged participants from Clemson University during the summer 2025. About two-thirds of the population were male and one-third were female.

Findings

The completion time and subjective workload of each task are distributed differently between Block A and Block B, especially in skewness.

		Task Incompletion Rate					
		D-Circuits	D-Folding	M-Legos	M-Screws	P-Block	P-Gear
Block 1		15.79%	47.83%	4.55%	0.00%	5.00%	9.09%
Block 2		4.35%	36.84%	0.00%	0.00%	0.00%	10.00%



Conclusion

Impact of mental fatigue can be detected with task completion time and incomplete rate.

Optimal Dynamic allocation models should consider the impact of task sequence and mental fatigue, as well as the training and skill levels of workers.

References and Acknowledgments

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